

Bowtie Reflection-Group 6

During the course of this bowtie diagram project, we began by learning the basis of bowtie, who can benefit from these types of visualizations, and different scenarios where risk assessment can be applied. We discovered that bowtie can be applied to everyday situations such as driving a car, the example used in this project, but that it can also be utilized by those in mechanical and industrial fields, along with many other practical applications. Through this, making sure that the example model is understandable, customizable, and shows the potential power behind bowtie influenced us in crafting our diagram.

We started by playing around with Reactflow Playground, adding nodes and connections, which enabled us to get an idea for what a more interactable bowtie would look like. The main problem with Playground was the lack of Java script and improving the interaction between nodes. We then contemplated using Approach 2 where we were considering making a more theoretical diagram whether in Playground or Canva, and we continued this direction until a team member figured out how to bring Github and Streamlit together to launch an app with our bowtie risk diagram.

Our main priority was emphasizing the customizable applications of bowtie and the interactive features that help to bring to life the bowtie diagram. Our audience was the potential clients. Risk assessment is undoubtedly top-of-mind for many of these companies, but figuring out a way to lay this out in a clear and actionable way to non-experts is possible through different means. Bowtie presents one way to provide this. So while our example is about driving a car, making sure to highlight ways in which it can be applied to many other problems was vital in building this project and ultimately convincing potential clients to buy into bowtie.

Overall, we deliberately kept the diagram more concise than the sample bowtie so it can be applied broadly across different business contexts and deliver a clearer, more universal message. Relatively smaller hazards may have fewer threats, prevention and migration barriers, and consequences, but we can also expand these to more robust hazards. We also created a “Fail Barrier” button for when a human or hardware intervention may not be available which leads to a failure downstream. It can easily be restored, but helps clients better visualize the effects of a failed barrier and plan effectively.

With a clear legend and deliberate color selection identifying each type of node, the diagram is clear to read at first glance. By clicking on each node, you are able to read more details, but reducing the number of words helps to reduce the sense of visual overload. One of the keys of any data visualization is decluttering, and we think we struck a good balance between showing the key aspects of bowtie while keeping distractions to a minimum. Once working with the client to build out their more polished bowtie, we would expect more wording and symbols to be present. However, as this is the mock bowtie to begin our meeting with the client, we find it more appealing, digestible and provides a strong foundation to build upon.

Overall, we hope that our project and the ideas behind them help Todus Advisors and you present bowtie diagrams for potential clients and their risk assessment needs. We believe that going to the basic tenants of data visualizations of decluttering and simplifying the message will help to both clarify and showcase the power of bowtie to potential clients going forward.